Javier Bonilla Cruz

Solar Energy Researcher

javier.bonilla@psa.es https://javier.solar LinkedIn GitHub
ORCID Google Scholar Scopus ResearchGate

Research Trajectory

I am a Senior Researcher in Concentrating Solar Power (CSP) and Renewable Energy Systems with over 15 years of experience in the modeling, simulation, control, and digitalization of large-scale energy infrastructures. My research focuses on the application of advanced automation, Artificial Intelligence (AI), and data-driven methodologies to improve the efficiency, flexibility, and sustainability of solar thermal power plants and hybrid renewable systems. I hold a PhD with International Mention and Extraordinary Prize in Engineering, awarded cum laude. During my doctoral research, including research stays at UNED Madrid and ETH Zürich, I developed pioneering switching moving boundary models for two-phase flow heat exhcnages in parabolic trough power plants. This work constituted the first modeling framework capable of representing all operating configurations.

My scientific output includes more than 30 peer-reviewed journal articles published mainly in high-impact international journals (Q1 and Q2), with a significant proportion ranked in the top citation percentiles of their respective fields. I am co-author of two scientific books. My research covers CSP plant modeling and control, thermal energy storage, heliostat field optimization, and the application of AI techniques such as deep learning, reinforcement learning, and digital twins. This AI–based research has attracted significant national and international media and outreach attention, including coverage by leading CSP and energy dissemination platforms (e.g., SolarPACES and Europa Press). My work has received best paper awards, as well as research prizes awarded to my research group. I have participated in 8 international and 10 national competitive research projects funded by Horizon Europe, Horizon 2020, FP7, and Spanish national programs, addressing next-generation CSP technologies, energy storage, hybrid renewable systems and digitalization. In several of these projects, I led and coordinated key tasks in modeling, control, and AI, and I have been actively involved in technology transfer activities, including a contract for the modeling and simulation of the commercial Andasol I parabolic trough solar power plant owned by Cobra Energy.

I founded CSP Data (https://cspdata.com), a free web-based platform providing comprehensive data on all commercial CSP plants worldwide. The platform supports researchers, policymakers, and investors by enabling data-driven analysis of trends, performance, and development opportunities. I am currently collaborating with RIFS Potsdam (Germany) to expand the platform with additional design and production data, with the objective of establishing CSP Data as a global reference for the sector. Closely related to this activity, I proposed a novel methodology and AI–based software tool to optimize electricity generation mixes using historical data. The tool minimizes production costs and curtailments while ensuring supply security and compliance with policy-driven decarbonization targets. Its intellectual property is registered under my sole authorship. Results for the Spanish electricity market are publicly available online (http://mixspain.psa.es). This technology was transferred through collaborations with the Portuguese Renewable Energy Association (APREN) and the Spanish Association for the Promotion of the Solar Thermal Industry (PROTERMOSOLAR) to assess the potential role of CSP in Portugal for the European Solar Thermal Electricity Association (ESTELA).

I have supervised two PhD students, both of whom obtained the maximum distinction and are currently postdoctoral researchers at CSIC and CIEMAT, with previous international experience at the Commonwealth Scientific and Industrial Research Organisation (CSIRO, Australia), Arizona State University (USA), and the German Aerospace Center (DLR, Germany). I have also supervised final degree projects and undergraduate research internships. I act as a reviewer for competitive R&D projects, including for the National Council of Science and Technology of Paraguay (CONACYT), and I participate in scientific committees and strategic research lines related to energy digitalization. I am an international program committee member of the International Renewable Energy Congress (IREC), within the IEEE Power & Energy Society, and a member of the joint programme on Digitalisation for Energy of the European Energy Research Alliance (EERA).

My activity includes a strong outreach and dissemination component, with participation in European Researchers' Nights, talks at high schools, outreach research projects, informative publications, blog posts, and the development of 3D-printed mock-ups and augmented reality models. I have developed open-source scientific software (e.g., Optifmus, Surf) and contributed to community-driven projects such as SolarPILOT and ThermoCycle. My full CV and supporting documentation are openly available at https://javier.solar.

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Education

Ph.D. with International Mention and Extraordinary Prize

University of Almería - 2008 - 2013

- **Topic**: Modeling of Two-Phase Flow Evaporators for Parabolic-Trough Solar Thermal Power Plants
- **Grade**: cum laude

Certificate | Extraordinary prize

Master in Advanced Computer Science

University of Almería - 2007 - 2008

- **Academic grade:** 3.54/4.
- **Project**: Approaches to the chattering problem in dynamic mathematical two-phase flow models
- **Grade**: honours.

Certificate | Academic record

Master in Computer Engineering

University of Almería - 2001 - 2004

- **Academic grade:** 2.54/4.
- **Project**: Decision making software for the climate conditions of tomato crops in greenhouses
- **Grade**: honours.

Certificate | Academic record | Qualifications framework

Bachelor in Computer Systems Engineering

University of Almería - 1998 - 2001

Official Program, Plan 1995, Escuela Superior de Ingeniería (ESI), Computing Department. **Grade**: 2.56/4.

Certificate | Academic record | Qualifications framework

Languages

- **English**: Advanced C1 level
- **Spanish**: Native

Certificado de nivel avanzado en el idioma inglés

2015

Escuela Oficial de Idiomas, Consejería de Educación de la Junta de Andalucía, July 2015

Certificate in Advanced English (CAE)

2010

University of Cambridge, ESOL Examinations, Effective Operational Proficiency Level, General English, Level C1 of CEFR. Grade: C, Score: 69/100, June 2010

Advanced English I

2010

Andalusian Entrepreneur Network Foundation (CEA), financed by the Andalusian Government and the European Social Fund, online course, 50 hours. Almería, July 2 - 17, 2010

First Certificate in English (FCE)

2009

University of Cambridge, ESOL Examinations, Vantage Level, General English, Level B2 of CEFR (The Common European Framework of Reference). Grade: B, Score: 75/100, June 2009

Work Experience

Work trajectory document - Informe de vida laboral

2004 - Now

Tesorería General de Seguridad Social, Ministerio de Inclusión, Seguridad Social y Migraciones, Gobierno de España

Attestation of positions - Certificado de puestos de trabajo

2026

CIEMAT-PSA, Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas - Plataforma Solar de Almería

Postdoctoral researcher - Permanent position

17/01/2024 - Now (2 años)

CIEMAT-PSA, Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas - Plataforma Solar de Almería

Postdoctoral researcher - SolarNOVA II project

16/04/2018 - 16/01/2024 (5 años, 9 meses)

CIEMAT-PSA

Postdoctoral researcher - EFFERDESAL project

19/12/2016 - 31/12/2017 (1 año, 12 días)

CIEMAT-PSA

Postdoctoral researcher - HYSOL project

30/10/2013 - 31/07/2016 (2 años, 9 meses)

CIEMAT-PSA

University graduate researcher - HIBIOSOLEO project

29/08/2011 - 28/09/2013 2 años, 1 mes)

CIEMAT-PSA

University graduate researcher - Economy and employment stimulation plan (Plan E)

01/07/2009 - 30/06/2011 (2 años)

CIEMAT-PSA

University graduate researcher - Work-experience contract

29/04/2009 - 30/06/2009 (2 meses)

CIEMAT-PSA

University graduate researcher in training - Ph.D. Scholarship

01/03/2007 - 28/04/2009 (2 años, 2 meses)

CIEMAT-PSA

Analyst - programmer

29/11/2004 - 16/02/2007 (2 años, 2 meses, 18 días)

Grupo J. Carrión, Asetir S.L., Volvo Industrial vehicles dealer (Veinsur), Huercal de Almería, (Almería)

Company scholarship, UAL - PSA agreement 2004

15/04/2004 - 31/10/2004 (6 meses, 16 días)

Reorganization of the Direct Steam Generation (DISS) in parabolic troughs project database, Plataforma Solar de Almería

Scientific Publications Overview

	Índice h	Citas	Docs.
Elsevier Scopus	15	698	44
Google Scholar	18	1100	69

Publicaciones JCR

D1	Q1	Q2	Q3	Q4	Total
12	19	10	2	2	33
36%	58%	30%	6%	6%	

Distribución de Citas (Google Scholar)

Primera Categoría

Solar Power Plants & Solar Thermal Energy

Scopus

Documents in top 25% journals

72.4% (21 documents)

Percent of documents in the top 25% journals by CiteScore

Scopus

Documents in top citation percentiles

38.7% (12 documents)

Percent of documents in the top 25% most cited documents worldwide

Scopus

Altmetric PLUMX

Mendeley	1597	Bluesky	3
Twitter (X)	27	Patentes	2
Noticias	24	Blogs	2
Facebook	11	Wikipedia	2

Dimensions

Field Citation Ratio (FCR)

4.95 veces más citas

Respecto a la media en el mismo área, categoría y año de publicación

Sustainable Development Goals

Goal 6: Clean water and sanitation	10 documents	Goal 13: Climate action	3 documents
Goal 7: Affordable and clean energy	34 documents	Goal 14: Life below water	1 document
Goal 9: Industry, innovation and infrastructure	1 document	Goal 15: Life on land	1 document

Scientific Publications (33)

Multimodal Deep Learning for Solar Radiation Forecasting

V. Abad-Alcaraz, M. Castilla / UAL, J.A. Carballo, **J. Bonilla** / CIEMAT, J.D. Álvarez / UAL - 2025

Applied Energy, Volume 393, ISSN: 1872-9118. DOI: 10.1016/j.apenergy.2025.126061, 2025.
JIF: 11 (2024),
Q1 - Energy & fuels (20/170, 89.3% percentile)
FWCI: 28.92 (OpenAlex)
Citation percentile (by year/subfield): 99.19% (OpenAlex)

A real-time and modular weather station software architecture based on microservices

J. Bonilla, J.A. Carballo / CIEMAT, V. Abad-Alcaraz, M. Castilla, J.D. Álvarez / UAL, J. Fernández-Reche / CIEMAT - 2025

Environmental Modelling & Software, 106337, ISSN: 1364-8152. DOI: 10.1016/j.envsoft.2025.106337, 2025.
JIF: 4.6 (2024),
Q2 - Computer science, interdisciplinary applications (50/175, 71.7% percentile)
Mendely readers: 36 (PlumX)

Reinforcement learning for heliostat aiming - Improving the performance of Solar Tower plants

J.A. Carballo, **J. Bonilla** / CIEMAT, N.C. Cruz / UAL, J. Fernández-Reche / CIEMAT, J.D. Álvarez / UAL, A. Avila-Marin / CIEMAT, M. Berenguel / UAL - 2025

Applied Energy, Volume 377, Part 8, ISSN: 1872-9118. DOI: 10.1016/j.apenergy.2024.124574, 2025.
JIF: 11 (2024),
Q1 - Energy & fuels (20/170, 89.3% percentile)
FWCI: 2.387 (OpenAlex)
Citation percentile (by year/subfield): 81.95% (OpenAlex)
Press news: 20 (Links)

Development of a novel tool to simulate solar thermal cogeneration plants using small-capacity tower plants

J. Bonilla, B. Ortega-Delgado, D.C. Alarcón-Padilla , J. Fernández-Reche, P. Palenzuela / CIEMAT - 2025

Energy Conversion and Management, Volume 324, 119285, ISSN: 0960-1481. DOI: 10.1016/j.enconman.2024.119285, 2025
JIF: 10.9 (2024),
Q1 - Mechanics (4/171, 98.0% percentile)

Modern SCADA for CSP Systems Based on OPC UA, WiFi Mesh Networks and Open Source Software

J.A. Carballo, **J. Bonilla**, J. Fernandez-Reche, A. Avila-Marin, B. Díaz / CIEMAT - 2024

Energies, Volume 17, Number 24, Article number 6284, ISSN: 1996-1073. DOI: 10.3390/en17246284, 2024.
JIF: 3.2 (2024),
Q3 - Energy & fuels (123/182, 32.69% percentile)
Mendely readers: 32 (PlumX)

Cloud Detection and Tracking Based on Object Detection with Convolutional Neural Networks

J.A. Carballo, **J. Bonilla**, J. Fernández-Reche / CIEMAT, B. Nouri / DLR, A. Avila-Marin / CIEMAT, Y. Fabel / DLR, D.C. Alarcón-Padilla / CIEMAT - 2023

Algorithms, Volume 16, Issue 10, 487, ISSN: 1999-4893.
DOI: 10.3390/a16100487, 2023.
JIF: 1.8 (2023),
Q2 - Computer science, theory and methods (62/144, 57.3% percentile)
FCR: 3.71 (Dimensions)

Dynamic modeling of a multi-effect vertical falling-film evaporator for water reuse in CSP plants

B. Ortega-Delgado, P. Palenzuela, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, L. Roca, D.C. Alarcón-Padilla / CIEMAT - 2022

Desalination, Volume 529, 115623, ISSN: 0011-9164. DOI: 10.1016/j.desal.2022.115623, 2022.
JIF: 9.9 (2022),
Q1 - Water resources (3/103, 98.06% percentile)

Feasibility and practical limits of full decarbonization of the electricity market with renewable energy: Application to the Spanish power sector

J. Bonilla, J. Blanco, E. Zarza, D.C. Alarcón-Padilla / CIEMAT - 2021

Energy, Volume 239, Part E, 122437, ISSN: 0360-5442. DOI: 10.1016/j.energy.2021.122437, 2021.
JIF: 8.857 (2021),
Q1 - Thermodynamics (3/63, 96.83% percentile)
FCR: 11.52 (Dimensions)

Optimal operation of solar thermal desalination systems coupled to double-effect absorption heat pumps

J.A. Carballo, **J. Bonilla**, L. Roca / CIEMAT, A. de la Calle / CSIRO, P. Palenzuela, D.C. Alarcón-Padilla / CIEMAT, M. Berenguel / UAL - 2020

Energy Conversion and Management, Volume 210, 112705, ISSN: 0960-1481. DOI: 10.1016/j.enconman.2020.112705, 2020
JIF: 9.709 (2020),
Q1 - Mechanics (2/135, 98.26% percentile)
Mendely readers: 34 (PlumX)

Solar tower power mockup for the assessment of advanced control techniques

J.A. Carballo, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, J. Fernández-Reche, G. García / CIEMAT - 2020

Renewable Energy, Volume 149, pages 682-690, ISSN: 0960-1481. DOI: 10.1016/j.renene.2019.12.075, 2020.
JIF: 8.001 (2020),
Q1 - Energy & fuels (16/144, 89.58% percentile)
Mendely readers: 48 (PlumX)

New approach for solar tracking systems based on computer vision, low cost hardware and deep learning

J.A. Carballo, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, J. Fernández-Reche, G. García / CIEMAT - 2019

Renewable Energy, Volume 133, pages 1158-1166, ISSN: 0960-1481. DOI: 10.1016/j.renene.2018.08.101, 2019.
JIF: 6.274 (2019),
Q1 - Energy & fuels (19/112, 83.93% percentile)
FCR: 20.6 (Dimensions)
Citation percentile (by year/subfield): 95.91% (OpenAlex)
Mendely readers: 157 (PlumX)

Parabolic trough collector field dynamic model: Validation, energetic and exergetic analyses

J.A. Carballo, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, P. Palenzuela / CIEMAT - 2019

Applied Thermal Engineering, Volume 148, pages 777-786, ISSN: 1359-4311. DOI: 10.1016/j.applthermaleng.2018.11.093, 2019.
JIF: 4.725 (2019),
Q1 - Engineering, mechanical (13/130, 90.77% percentile)
FCR: 3.53 (Dimensions)

Design and experimental validation of a computational effective dynamic thermal energy storage tank model

J. Bonilla, M. M. Rodríguez-García, L. Roca / CIEMAT, A. de la Calle / CSIRO, L. Valenzuela / CIEMAT - 2018

Energy, Volume 152, pages 840-857, ISSN: 0360-5442. DOI: 10.1016/j.energy.2017.11.017, 2018.
JIF: 5.537 (2018),
Q1 - Thermodynamics (3/60, 96.67% percentile)
FCR: 2.96 (Dimensions)

A novel heat exchanger design method using a delayed rejection adaptive Metropolis Hasting algorithm

A. Mohammadi / UT, **J. Bonilla** / CIEMAT, R. Zarghami, S. Golshan / UT - 2018

Applied Thermal Engineering, Volume 137, pages 808-821, ISSN: 1359-4311. DOI: 10.1016/j.applthermaleng.2018.04.028, 2018.
JIF: 4.026 (2018),
Q1 - Engineering, mechanical (13/129, 90.69% percentile)

Model validation and control strategy of a heat recovery system integrated in a renewable hybrid power plant demonstrator

J. Bonilla, L. Roca / CIEMAT - 2018

Solar Energy, Volume 176, pages 698-708, ISSN: 1099-1514. DOI: 10.1016/j.solener.2018.10.076, 2018.
JIF: 4.674 (2018),
Q2 - Energy & fuels (39/131, 70.99% percentile)

Steady-state and dynamic validation of a parabolic trough collector model using the ThermoCycle Modelica library

A. Desideri, R. Dickes / ULG, **J. Bonilla**, L. Valenzuela / CIEMAT, S. Quoilin, V. Lemort / ULG - 2018

Solar Energy, Volume 174, pages 866-877, ISSN: 1099-1514. DOI: 10.1016/j.solener.2018.08.026, 2018.
JIF: 4.674 (2018),
Q2 - Energy & fuels (39/131, 70.99% percentile)
FCR: 5.46 (Dimensions)

New low-cost solar tracking system based on open source hardware for educational purposes

J.A. Carballo, **J. Bonilla**, L. Roca / CIEMAT, M. Berenguel / UAL - 2018

Solar Energy, Volume 164, pages 826-836, ISSN: 1099-1514. DOI:10.1016/j.solener.2018.09.064, 2018.
JIF: 4.674 (2018),
Q2 - Energy & fuels (39/131, 70.99% percentile)

Optimal operating conditions analysis for a multi-effect distillation plant according to energetic and exergetic criteria

J.A. Carballo / UAL, **J. Bonilla**, L. Roca / CIEMAT, A. de la Calle / CSIRO, P. Palenzuela, D.C. Alarcón-Padilla / CIEMAT - 2018

Desalination, Volume 435, pages 70-76, ISSN: 0011-9164. DOI: 10.1016/j.desal.2017.12.013, 2018.
JIF: 6.035 (2018),
Q1 - Water resources (2/91, 98.90% percentile)

Study on shell-and-tube heat exchanger models with different degree of complexity for process simulation and control design

J. Bonilla / CIEMAT, A. de la Calle / CSIRO, L. Roca / CIEMAT, M. M. Rodríguez-García, L. Valenzuela / CIEMAT, - 2017

Applied Thermal Engineering, Volume 124, pages 1425-1440, ISSN: 1359-4311. DOI: 10.1016/j.applthermaleng.2017.06.129, 2017.

JIF: 3.771 (2017),
Q1 - Engineering, mechanical (12/128, 91.41% percentile)
FCR: 8.46 (Dimensions)
Citation percentile (by year/subfield): 90.28% (OpenAlex)

Optimal operating conditions analysis of a multi-effect distillation plant

A. Carballo / UAL, **J. Bonilla**, L. Roca / CIEMAT, A. de la Calle / CSIRO, P. Palenzuela / CIEMAT, M. Berenguel / UAL - 2017

Desalination and Water Treatment, 69, pages 229-235, ISSN: 1944-3986. DOI: 10.5004/dwt.2017.0703, 2017.
JIF: 1.383 (2017),
Q3 - Engineering, chemical (79/137, 43.07% percentile)

Modelo dinámico de un recuperador de gases - sales fundidas para una planta termosolar híbrida de energías renovables

J. Bonilla, L. Roca / CIEMAT, A. de la Calle / CSIRO, S. Dormido / UNED - 2017

RIAI - Revista Iberoamericana de Automatica e Informatica Industrial, Volume 14, Issue 1, pages 70-81, ISSN: 1697-7912. DOI: 10.1016/j.riai.2016.11.003.
JIF: 0.494 (2017),
Q4 - Automation & control systems (60/61, 3.27% percentile)

Dynamic modeling and simulation of a double-effect absorption heat pump

A. de la Calle / CSIRO, L. Roca, **J. Bonilla**, P. Palenzuela / CIEMAT - 2016

International Journal of Refrigeration, Volume 72, pages 171-191, ISSN: 1877-0509. DOI: 10.1016/j.ijrefrig.2016.07.018, 2016.
JIF: 2.779 (2016),
Q1 - Thermodynamics (12/58, 81.03% percentile)

Predictive control applied to a solar desalination plant connected to a greenhouse with daily variation of irrigation water demand

L. Roca / CIEMAT, J. Sanchez, F. Rodríguez / UAL, **J. Bonilla** / CIEMAT, A. de la Calle / CSIRO, M. Berenguel / UAL - 2016

Energies, Volume 9, Number 3, Article number 194, ISSN: 1996-1073. DOI: 10.3390/en9030194, 2016.
JIF: 2.264 (2016),
Q2 - Energy & fuels (45/92, 52.17% percentile)

Switching moving boundary models for two-phase flow evaporators and condensers

J. Bonilla / CIEMAT, S. Dormido / UNED, F.E. Cellier / ETH - 2015

Communications in Nonlinear Science and Numerical Simulation, Volume 20, Issue 3, pages 743-768, ISSN: 1007-5704. DOI: 10.1016/j.cnsns.2014.06.035, 2015.
JIF: 2.834 (2015),
Q1 - Mathematics, applied (5/254, 98.43% percentile)
FCR: 14.64 (Dimensions)
Citation percentile (by year/subfield): 96.52% (OpenAlex)

Dynamic modeling and simulation of a solar-assisted multi-effect distillation plant

A. de la Calle, **J. Bonilla**, L. Roca, P. Palenzuela / CIEMAT - 2015

Desalination, Volume 357, pages 65-76, ISSN: 0011-9164. DOI: 10.1016/j.desal.2014.11.008, 2015.
JIF: 4.412 (2015),
Q1 - Water resources (2/85, 98.82% percentile)

Dynamic modeling and performance of the first cell of a multi-effect distillation plant

A. de la Calle, **J. Bonilla**, L. Roca / CIEMAT, P. Palenzuela / UTA - 2015

Applied Thermal Engineering, Volume 70, Issue 1, pages 410-420, ISSN: 1359-4311. DOI: 10.1016/j.applthermaleng.2014.05.035, 2015.
JIF: 2.739 (2014),
Q1 - Engineering, mechanical (8/130, 94.62% percentile)

A control based on a Knapsack problem for solar hydrogen production

L. Roca, R. Díaz, A. de la Calle, **J. Bonilla**, A. Vidal / CIEMAT - 2014

Optimal Control, Applications and Methods - Optimal Control of Solar Energy Systems (special issue), Volume 37, Issue 3, pages 496-507, ISSN: 1099-1514. DOI: 10.1002/oca.2118, 2014.
JIF: 0.903 (2014),
Q2 - Mathematics, applied (110/257, 57.59% percentile)

A room simulation tool for thermal comfort control in a bioclimatic building. A real example of use with an optimal controller

M. Castilla / UAL, **J. Bonilla** / CIEMAT, J.D. Álvarez / US, F. Rodríguez / UAL - 2014

Optimal Control, Applications and Methods - Optimal Control of Solar Energy Systems (special issue), Volume 37, Issue 3, pages 479-495, ISSN: 1099-1514. DOI: 10.1002/oca.2116, 2014.
JIF: 0.903 (2014),
Q2 - Mathematics, applied (110/257, 57.59% percentile)

Chattering in dynamic mathematical two-phase flow models

J. Bonilla, L.J. Yebra / CIEMAT, S. Dormido / UNED - 2012

Applied Mathematical Modelling, Volume 36, Issue 5, pages 2067-2081, ISSN: 0307-904X. DOI: 10.1016/j.apm.2011.08.013, 2012.
JIF: 1.706 (2012),
Q1 - Engineering, multidisciplinary (11/90, 88.89% percentile)

Parabolic-trough solar thermal power plant simulation scheme, multi-objective genetic algorithm calibration and validation

J. Bonilla, L.J. Yebra / CIEMAT,S. Dormido / UNED, E. Zarza / CIEMAT - 2012

Solar Energy, Volume 86, Issue 1, pages 531-540, ISSN: 0038-092X. DOI: 10.1016/j.solener.2011.10.025, 2012.
JIF: 2.952 (2012),
Q2 - Energy & fuels (21/81, 75.31% percentile)
FCR: 12.09 (Dimensions)
Citation percentile (by year/subfield): 78.75% (OpenAlex)

A heuristic method to minimise the chattering problem in dynamic mathematical two-phase flow models

J. Bonilla, L.J. Yebra / CIEMAT, S. Dormido / UNED - 2011

Mathematical and Computer Modelling, Volume 54, Issue 5-6, pages 1549-1560, ISSN: 0895-7177. DOI: 10.1016/j.mcm.2011.04.026, 2011.
JIF: 1.346 (2011),
Q1 - Computer science, software engineering (24/104, 77.88% percentile)

Mean densities in dynamic mathematical two-phase flow models

J. Bonilla, L.J. Yebra / CIEMAT, S. Dormido / UNED - 2010

Computer Modeling in Engineering & Science, Volume 66, Issue 1, pages 13-37, ISSN: 1526-1506. DOI: 10.3970/cmes.2010.067.013, 2011.
JIF: 1.123 (2010),
Q2 - Engineering, multidisciplinary (23/90, 75.56% percentile)

Object oriented modelling and simulation of ACUREX solar thermal power plant

L.J. Yebra / CIEMAT, M. Berenguel / UAL, **J. Bonilla** / CIEMAT, L. Roca / CIEMAT, S. Dormido / UNED, E. Zarza / CIEMAT - 2010

Mathematical and Computer Modelling of Dynamical Systems, Volume 16, Issue 3, pages 211-224, ISSN: 1744-5051. DOI: 10.1080/13873954.2010.507420, 2010.
JIF: 0.452 (2010),
Q4 - Mathematics, applied (191/236, 19.49% percentile)

International Research Projects (8)

I have participated as a researcher in the following internationally funded research projects.

Leading-edge cooperative advances towards the next generation of concentrated solar power (CSP) technology (COOPERANT)

2024 - 2028

Horizon Europe Framework Programme (HORIZON) Proposal number: 101172882, Researcher, budget: 2,999,500 €
COOPERANT website

Air-based solar thermal electricity for efficient renewable energy integration & compressed air energy storage (ASTERIx-CAESar)

2023 - 2027

Horizon Europe Framework Programme (HORIZON), HORIZON Innovation Actions, Innovative components and/or sub-systems for CSP plants and/or concentrating solar thermal installations, HORIZON-CL5-2022-D3-03-01, proposal number 101122231, Researcher, budget: 5,498,425 €
ASTERIx-CAESar website

Sustainable Near-net-shape Fabrication of Low Environmental impact Receiver materials (SUNFLOWER)

2023 - 2026

Clean Energy Transition Partnership Joint call 2022 (CETP 2022), European Union Proposal number: PCI2023-145996-2, Researcher, budget: 1,437,670 €
SUNFLOWER website

soLar field mEasurements to Increase performAnce (LEIA)

2022 - 2025

Horizon Europe Framework Programme (HORIZON), CSP ERANET Additional Call 2021 Proposal number: PCI2022-135015-2, Researcher, budget: 2,692,989 €
LEIA website

Expansion, improvement, renovation and remodelling of Plataforma Solar de Almería (PSA) (SolarNOVA II)

2018 - 2023

European Regional Development Fund (ERDF) and Spanish funds, ICTS-2017-03-CIEMAT-04, Researcher, budget: 11,996,375 €
SolarNOVA website

Solving water issues for CSP plants (SOLWARIS)

2018 - 2022

Horizon 2020, European Commission, grant agreement no 792103, Researcher, budget: 10,812,504 €
SOLWARIS website

Innovative configuration for a fully renewable hybrid CSP plant (HYSOL)

2013 - 2016

EU Seventh Framework Programme for Research (FP7-Energy), European Commission, grant agreement n° 308912, Researcher, budget: 9,275,982 €
HYSOL website

Solar facilities for the european research area (SFERA)

2009 - 2013

EU Seventh Framework Programme for Research (FP7-Infrastructures), European Commission, grant agreement n° 228296, Work team member, budget: 9,042,935 €
SFERA website

National Research Projects (10)

I have contributed as a researcher to the following funded rnational research projects.

More efficient heliostat fields for solar tower plants (HELIOSUN)

2022 - 2025

Generation Knowledge Projects, Ministry of Science and Innovation, PID2021-126805OB-100, Work team member, budget: 145,200 €

COntrol and ManageMent systems using Information and communications Technologies FOR ZERO Energy Buildings (COMMIT4.0EB)

2022 - 2025

Generation Knowledge Projects, Ministry of Science and Innovation, PID2021-126889OB-I00, Work team member, budget: 125,235 €
COMMIT4.0EB website

Hybrid cooling solutions for water saving in solar thermal applications (SOLHYCOOL)

2022 - 2025

Generation Knowledge Projects, Ministry of Science and Innovation, PID2021-126452OA-I00, Work team member, budget: 78,650 €

New Technologies for Enhancing Energy Efficency in Buildings (Ntech4Build)

2022 - 2024

Grant TED2021-131655B-I00 funded by AEI/10.13039/501100011033 and by the European Union NextGenerationEU, Work team member, budget: 168,590 €
Ntech4Build website

SOL-préndete: Didáctica y divulgación de la energía solar térmica de concentración con nuevas tecnologías de realidad aumentada y virtual

2022 - 2023

Convocatoria de ayudas para el Fomento de la Cultura Científica, Tecnológica y de la Innovación, Ministry of Science and Innovation, FCT-21-16905, Work team member, budget: 15.000 €
SOL-préndete website

Solar thermal solutions for integration in industrial processes (SOLTERMIN)

2018 - 2022

Spanish National Programme for Research Aimed at the Challenges of Society, Ministry of Science and Innovation, ENE2017-83973-R, Work team member, budget: 278,300 €
SOLTERMIN website

Control and energy management strategies in production environments with support of renewable energy (ENERPRO) – Subproject title: Efficient energy control and management of solar thermal desalination systems (EFFERDESAL)

2015 - 2018

Spanish National Programme for Research Aimed at the Challenges of Society, Ministry of Science and Innovation, DPI2014-56364-C2-2-R, Work team member, budget: 131,200 €
ENERPRO website

Controlling the growth of crops under glass by optimizing sustainability, economic and energy efficiency criteria (CONTROLCROP)

2011 - 2015

Proyecto de investigación de Excelencia financiado por la Consejería de Innovación Ciencia y Empresa, Junta de Andalucía, P10-TEP-6174, Work team member, budget: 145,005 €
CONTROLCROP website

Predictive control techniques for efficient management of renewable energy microgrids (POWER)

2011 - 2014

Programa nacional de proyectos de investigación fundamental, subprograma de proyectos de investigación fundamental no orientada, Ministry of Science and Innovation, DPI2010-21589-C05-04, Work team member, budget: 105,270 €
POWER website

Hybridization of renewable technologies in a power generation plant (HIBIOSOLEO)

2011 - 2013

INNPACTO 2010, National Program for Public-Private Cooperation, Ministry of Science and Innovation, IPT-440000-2010-004, Work team member, budget: 4,270,000 €
HIBIOSOLEO website

Books

Artificial Intelligence Applied to Solar Thermal Power Plants (EN PREPARACIÓN)

2027

V. Abad-Alcaraz, M. Leal, J.D. Álvarez, **J. Bonilla**, J.A. Carballo, M. Castilla Springer Nature.

Modelado y simulación dinámica de procesos termoquímicos en instalaciones termosolares

A. de la Calle, L. Roca, **J. Bonilla**, S. Dormido - 2015

Editorial CIEMAT, ISBN: 978-84-7834-743-8.
Depósito Legal: M-32433-2015, NIPO: 721-15-053-4.

Modeling and simulation of two-phase flow evaporators for parabolic-trough solar thermal power plants

J. Bonilla, L. J. Yebra, S. Dormido, E. Zarza - 2013

Editorial CIEMAT, ISBN: 978-84-7834-705-6.
Depósito Legal: M-28846-2013, NIPO: 721-13-044-9.

Invited Presentations (8)

Optimizing the Electricity Mix with AI: Application to the Spanish Grid

J. Bonilla / CIEMAT - 2025

Future Energy Mix with 100% Decarbonization Level, Online R&I Workshop, CST4ALL, Oral presentation, March 24, 2025.

Energy: Experience on Artificial Intelligence (AI)

J. Bonilla / CIEMAT - 2023

Workshop on the application of artificial intelligence at CIEMAT's projects, Oral presentation, online, July 3, 2023.

Assessment of the integration of a MED-TVC plant into a solar tower with Brayton cycle

J. Bonilla, P. Palenzuela, B. Ortega-Delgado, J.M. Serrano, D.C. Alarcón-Padilla / CIEMAT - 2022

Workshop on Renewable Energies and Energy Efficiency in water treatments, EERES4WATER Project, September 26, 2022.

The potential of concentrating solar power in portugal using inductive projection planning

J. Blanco, **J. Bonilla** / CIEMAT, G. Martín / Protermosolar - 2022

How CSP can improve Frequency Stability of the Power Grid, ATA insights, Oral presentation, online, January 27, 2022.

Optimization of CSP development in spain using inductive projection planning tool based on artificial intelligence

J. Blanco, **J. Bonilla**, E. Zarza, D.C. Alarcón-Padilla / CIEMAT - 2020

CSP4Climate, The Cyprus Institute, Oral presentation, online, December 15 - 17, 2020

Herramienta para optimizar el mix eléctrico aplicando técnicas de inteligencia artificial

J. Bonilla / CIEMAT - 2020

Feria Internacional de Energía y Medioambiente (GENERA), Oral presentation, Madrid (Spain), Feb 5 - 7, 2020

Modelado y simulación de plantas termosolares

J. Bonilla / CIEMAT - 2014

Jornadas de Informática 2014, University of Almería, Oral presentation, Almería (Spain), March 3, 2014

Modelado y simulación de plantas termosolares

J. Bonilla / CIEMAT - 2010

XXXI Jornadas de Automática 2010, Modeling and Simulation Group, Oral presentation, Jaén (Spain), September 8, 2010

Conferences Overview

Congresos	Presentaciones	Póster	Total
Internacionales	24 (69%)	11	35
Nacionales	1	9	10

	Citas
Elsevier Scopus	76
Google Scholar	132

Altmetric

Mendeley 353

International Conference Proceedings (35)

Digital tools and artificial intelligence for the optimization of central receiver tower systems (ACCEPTED)

J. Bonilla / CIEMAT, J.Carballo, A. Ávila-Marín / CIEMAT, M. Berenguel /UAL - 2026

XX Congreso Ibérico y XVI Congreso Iberoamericano de Energía Solar Abstract, oral presentation, Almería (Spain), Mar 23 - 25, 2026.

Optimization of heliostat aiming strategies in tower power plants using reinforcement learning (ACCEPTED)

J.Carballo, **J. Bonilla** / CIEMAT, N.C. Cruz / UGR, J.D.Álvarez / UAL, A. Ávila-Marín / CIEMAT, M. Berenguel /UAL - 2026

XX Congreso Ibérico y XVI Congreso Iberoamericano de Energía Solar Abstract, oral presentation, Almería (Spain), Mar 23 - 25, 2026.

Integration of solar radiation nowcasting models into a real-time modular weather station architecture

J. Bonilla, J.A. Carballo / CIEMAT, V. Abad-Alcaraz, M. Castilla, J.D. Álvarez / UAL, J. Fernández-Reche / CIEMAT - 2025

31st Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, oral presentation, Almería (Spain), Sep 23 - 26, 2025.

Advancing Heliostat Aiming Strategies in Solar Tower Plants

J.A. Carballo, **J. Bonilla** / CIEMAT, N.C. Cruz / UGR, J. Fernández-Reche / CIEMAT, J.D. Álvarez / UAL, A. Avila-Marin / CIEMAT, M. Berenguel / UAL - 2025

31st Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, Poster, Almería (Spain), Sep 23 - 26, 2025.

More Efficient Heliostat Fields for Solar Tower Plants: The HELIOSUN Project

J. Ballestrín, L. Valenzuela, J. Fernández-Reche / CIEMAT, R. Pujol, G. Cardona / UIB, J.A. Carballo, **J. Bonilla**, N. Estremera, E. Carra, N. Simal, R. Monterreal / CIEMAT, A. Marzo / UGR - 2024

30th Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, Poster, Rome (Italy), Oct 8 - 11, 2024.

Desarrollo de modelos de predicción de la radiación solar directa basados en redes neuronales mediante el uso de imágenes

V. Abad, M. Castilla, J.D. Álvarez / UAL, J.A. Carballo, **J. Bonilla** / CIEMAT - 2024

Regular paper in Proceedings, Oral presentation
CIES, XIX Congreso Ibérico y XV Congreso Iberoamericano de Energía Solar Évora (Portugal), Jun 19 - 21, 2024

Optimal operation of a combined cooling system

J.M. Serrano / CIEMAT, J.D. Gil / UAL, **J. Bonilla**, P. Palenzuela, L. Roca / CIEMAT - 2024

Regular paper in Proceedings, Oral presentation
4th IFAC Conference on Advances in Proportional-Integral-Derivative Control, Almería (Spain), Jun 12 - 14, 2024

Strategies for enhancing the use of solar energy in utility-scale desalination plants

G. Zaragoza, D.C. Alarcón-Padilla, **J. Bonilla**, P. Palenzuela / CIEMAT - 2023

Regular paper in Proceedings, Oral presentation
The International Association Desalination (IDA) Seville summit on Water and Climate Change, Seville (Spain), Oct 15 - 18, 2023

CSP Data: a data discovery web application of commercial CSP plants

J. Bonilla / CIEMAT, R. Thoning / RIFS, J. Carballo / CIEMAT, J. Lilliestam / RIFS, D.C. Alarcón-Padilla, E.Zarza / CIEMAT - 2023

29th Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, Poster, DOI: 10.52825/solarpaces.v1i.721, Sydney (Australia), Oct 10 - 13, 2023.

Smart Heliostat Tracking System based on Artificial Intelligence

J.A. Carballo, **J. Bonilla**, J. Fernández-Reche, J. Ballestrín, L. Valenzuela / CIEMAT - 2023

29th Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, oral presentation, Sydney (Australia), Oct 10 - 13, 2023.

New Control Architecture for the Next Generation of Central Receiver Power Plants

J.A. Carballo, J. Fernández-Reche, **J. Bonilla**, A. Ávila / CIEMAT - 2023

29th Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, Poster, Sydney (Australia), Oct 10 - 13, 2023.

On the use of artificial neural networks for automatic heliostat aiming

J.A. Carballo / CIEMAT, N.C. Cruz / UAL, **J. Bonilla** / CIEMAT, J.D. Álvarez and M. Berenguel / UAL - 2023

Regular paper in Proceedings, Oral presentation
17th International Work-Conference on Artificial Neural Networks (IWANN)
Ponta Delgada, Azores (Portugal), Jun 19 - 21, 2023

Hel-IoT web server: a smart heliostat development platform

J.A. Carballo, **J. Bonilla**, J. Fernández-Reche, D.C. Alarcón-Padilla / CIEMAT - 2022

28th Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, Poster, DOI: 10.52825/solarpaces.v1i.721, Albuquerque (USA), Sep 27 - 30, 2022.

Assessment of the integration of a MED-TVC plant into a solar tower with Brayton cycle

J. Bonilla, P. Palenzuela, B. Ortega-Delgado, J.M. Serrano, D.C. Alarcón-Padilla / CIEMAT - 2022

Desalination for the environment: clean water and energy,
Abstract, Poster, Las Palmas (Spain), Jun 20 - 23, 2021.

Inductive projection planning: Putting CSP in the picture

J. Bonilla / CIEMAT, L. Crespo / Protermosolar - 2020

26th Concentrating Solar Power and Chemical Energy Systems (SolarPaces),
Regular paper in Proceedings, Oral presentation, DOI: 10.1063/5.0085735,
Albuquerque (USA), Online, Sep 28 - Oct 2, 2020.

Optimización del desarrollo de la energia termosolar en españa en el contexto de la transición energética

J. Blanco, **J. Bonilla**, E. Zarza, D.C. Alarcón-Padilla / CIEMAT - 2020

XVII Congreso Ibérico e XIII Congreso Ibero-americano de Energia Solar (CIES),
Regular paper in Proceedings, Oral presentation, DOI: 10.34637/cies2020.1.1003,
Lisboa (Portugal), Online, November, 3 - 5, 2020.

Machine learning perspectives in concentrating solar thermal technology

J. Bonilla, J.A. Carballo / CIEMAT, M. Berenguel / UAL, J. Fernández-Reche, L. Valenzuela / CIEMAT - 2019

10th Congress of the Federation of European Simulation Societies (EUROSIM).
Special Session on Trends and Perspectives of Machine Learning in Automation,
Regular paper in Proceedings, Oral presentation, La Rioja (Spain), Jul 1 - 5, 2019.

A software for dimensioning of small microgrids with PV-battery systems

J.L. Torres-Moreno / UAL, J.A. Carballo, **J. Bonilla** / CIEMAT, A. Gímenez / UAL - 2019

ISES Solar World Congress, IEA SHC International Conference on Solar Heating and Cooling for Buildings and Industry, Regular paper in Proceedings, Oral presentation, DOI: 10.18086/swc.2019.31.05.
Santiago (Chile), Nov 4 - 7, 2019.

Machine learning for solar trackers

J.A. Carballo, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, J. Fernández-Reche, G. García / CIEMAT - 2018

24th Concentrating Solar Power and Chemical Energy Systems (SolarPaces), Regular paper in Proceedings, Poster, DOI: 10.1063/1.5117524, Casablanca (Morocco), Oct 2 - 5, 2018.

Development of an open source multi-platform software tool for parameter estimation studies in FMI models

J. Bonilla, J.A. Carballo, L. Roca / CIEMAT, M. Berenguel / UAL - 2017

12th International Modelica conference, Regular paper in Proceedings, Oral presentation,
DOI: 10.3384/ecp17132683, Prague (Czech Republic), May 15 - 17, 2017.

Optimal operating conditions analysis for a double-effect absorption heat pump coupled to a multi effect distillation plant

J.A. Carballo, **J. Bonilla**, P. Palenzuela, L. Roca / CIEMAT, A. de la Calle / CSIRO - 2017

Euromed, Desalination for Clean Water and Energy: Cooperation around the World, Regular paper in Proceedings, Poster, Tel Aviv (Israel), May 9 - 12, 2017.

Control strategies applied in HYSOL demonstrator: a simulation-based evaluation

L. Roca, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, L. González, A.R. Rocha / COBRA - 2016

24th Mediterranean Conference on Control and Automation, Regular paper in Proceedings, Oral presentation,
DOI: 10.1109/MED.2016.7535952, Athens (Greece), June 21 - 24, 2016.

Operation & training tool for a gas - molten salt heat recovery demonstrator facility

J. Bonilla, L. Roca / CIEMAT, E. Cerrajero, D. Lopez / IDIE, S. Miraval, S. Padilla / AITESA, L. Diez / SERLED, A.R. Rocha, L. González / COBRA - 2016

6th International Conference on Sustainable Energy Information Technology (SEIT-2016), Regular paper in Proceedings, Oral presentation, DOI: 10.1016/j.procs.2016.04.232, Madrid (Spain), May 23 - 26, 2016.

Optimal operating conditions analysis of a multi-effect distillation plant

J.A. Carballo, **J. Bonilla**, L. Roca / CIEMAT, A. de la Calle / CSIRO, P. Palenzuela / CIEMAT, Manuel Berenguel / UAL - 2016

Desalination for the environment: clean water and energy, Abstract, Poster, Rome (Italy), May 22 - 26, 2016.

Object-oriented modeling of a multi-pass shell-and-tube heat exchanger and its application to performance evaluation

J. Bonilla, M.M. Rodríguez, L. Roca, L. Valenzuela / CIEMAT - 2015

1st Conference on Modelling, Identification and Control of Nonlinear Systems (MICNON), Regular paper in Proceedings, Oral presentation, DOI: 10.1016/j.ifacol.2015.09.166, Saint-Petersburg (Rusia), June 24 - 26, 2015.

Control strategies in a thermal oil - molten salt heat exchanger

L. Roca, **J. Bonilla**, M.M. Rodríguez, P. Palenzuela, A. de la Calle, L. Valenzuela / CIEMAT - 2015

21th Concentrating Solar Power and Chemical Energy Systems (SolarPaces), Regular paper in Proceedings, Poster, DOI: 10.1063/1.4949227, Cape Town (South Africa), October 13 - 16, 2015.

Experimental calibration of heat transfer and thermal losses in a shell-and-tube heat exchanger

J. Bonilla, A. de la Calle, M.M. Rodríguez, L. Roca, L. Valenzuela / CIEMAT - 2015

11th International Modelica Conference, Regular paper in Proceedings, Poster, DOI: 10.3384/ecp15118873, Versailles (France), September 21 -23, 2015.

A combinatorial optimization problem to control a solar reactor

L. Roca, R. Diaz-Franco, A. de la Calle, **J. Bonilla**, L. J. Yebra, Alfonso Vidal / CIEMAT - 2013

19th Concentrating Solar Power and Chemical Energy Systems (SolarPaces), Regular paper in Proceedings, Oral presentation, DOI: 10.1016/j.egypro.2014.03.216, Las Vegas (USA), September 17 - 20, 2013.

Object-oriented library of switching moving boundary models for two-phase flow evaporators and condensers

J. Bonilla, L.J. Yebra / CIEMAT, S. Dormido / UNED, F.E. Cellier / ETH - 2012

9th International Modelica Conference, Regular paper in Proceedings, Oral presentation, DOI: 10.3384/ecp1207671, Munich (Germany), September 3 - 5, 2012.

Object-oriented modeling of switching moving boundary models for two-phase flow evaporators

J. Bonilla, L. J. Yebra /CIEMAT, S. Dormido / UNED, F. E. Cellier / ETH - 2012

7th Vienna International Conference on Mathematical Modelling (MATHMOD), Regular paper in Proceedings, Oral presentation, DOI: 10.3182/20120215-3-AT-3016.00189, Vienna (Austria), February 5 - 7, 2012.

Exploiting OpenMP in the initial section of modelica models

J. Bonilla, L.J. Yebra / CIEMAT, S. Dormido / UNED - 2011

4th International Workshop on Equation-Based Object-Oriented Modeling Languages and Tools, Regular paper in Proceedings, Oral presentation, Zürich (Switzerland), September 5, 2011.

Real-time simulation of CESA-I central receiver solar thermal power plant

J. Bonilla, L. Roca, L.J. Yebra / CIEMAT, S. Dormido / UNED - 2009

7th International Modelica Conference, Regular paper in Proceedings, Oral presentation, DOI: 10.3384/ecp09430062, Como (Italy), September 20 - 22, 2009.

Chattering in dynamic mathematical two-phase flow models

J. Bonilla, L.J. Yebra , E. Zarza / CIEMAT, S. Dormido /UNED - 2009

European Control Conference 2009 (ECC), Regular paper in Proceedings, Oral presentation, DOI: 10.23919/ECC.2009.7075015, Budapest (Hungary), August 23 - 26, 2009.

Object oriented modelling and simulation of acurex solar thermal power plant

L.J. Yebra / CIEMAT, M. Berenguel / UAL, L. Roca / Aunergy Thermosolar S.L., **J. Bonilla** /CIEMAT, S. Dormido / UNED - 2009

6th Vienna International Conference on Mathematical Modelling (MATHMOD), Extended Abstract and Regular paper in Proceedings, Oral presentation, Vienna (Austria), February 11 - 13, 2009.

Modelling and real-time simulation of heliostat fields in Central Receiver Plants

J. Bonilla /CIEMAT, L. Roca, J. González / Aunergy Thermosolar S.L., L.J. Yebra / CIEMAT - 2009

6th Vienna International Conference on Mathematical Modelling (MATHMOD), Extended Abstract and Short paper in Proceedings, Poster, Vienna (Austria), February 11 - 13, 2009.

National Conference Proceedings (10)

Desarrollo de modelos de predicción de radiación solar mediante técnicas de machine learning

V. Abad-Alcaraz, M.M. Castilla, J.D. Álvarez / UAL, J.A. Carballo, **J. Bonilla** / CIEMAT, - 2023

XLIV Jornadas de Automática 2023. Regular paper in Proceedings, Poster, DOI: 10.17979/spudc.9788497498609.382, Zaragoza (Spain), September 6 - 8, 2023.

Operación óptima de un sistema de refrigeración combinada

J.M. Serrano / CIEMAT, J.D. Gil / UAL, **J. Bonilla**, P. Palenzuela, L.Roca / CIEMAT - 2023

XLIV Jornadas de Automática 2023. Regular paper in Proceedings, Poster, DOI: 10.17979/spudc.9788497498609.477, Zaragoza (Spain), September 6 - 8, 2023.

Control de un recuperador de sales en una planta termosolar híbrida

P. Otalóra / UAL, L. Roca, **J. Bonilla** / CIEMAT, J.L. Gúzman / UAL - 2019

XL Jornadas de Automática 2019. Regular paper in Proceedings, Poster, DOI: 10.17979/spudc.9788497497169.421, Ferrol (Spain), September 4 - 6, 2019.

Modelado y optimización para una gestión eficiente de recursos en desalación solar

J.A. Carballo, **J. Bonilla** / CIEMAT, M. Berenguel / UAL - 2018

XVI Simposio CEA de Ingeniería de Control, Regular paper in Proceedings, Poster, Almería (Spain), March 7 - 9, 2018.

Control predictivo para satisfacer la demanda de agua en un invernadero mediante un sistema de desalación solar

L. Roca / CIEMAT, J.A. Sanchez, F. Rodríguez / UAL, **J. Bonilla** / CIEMAT, A. de la Calle / CSIRO, M. Berenguel / UAL, - 2016

II Simposio Nacional de Ingeniería Hortícola - Automatización y TICs en Agricultura, Regular paper in Proceedings, Poster, Almería (Spain), February 10 - 12, 2016

Inversión de la causalidad computacional en el modelado dinámico. Caso práctico de una planta de generación de hidrógeno solar

A. de la Calle, L. Roca, **J. Bonilla** / CIEMAT - 2015

XXXVI Jornadas de Automática 2015, Regular paper in Proceedings, Poster, Bilbao (Spain), September 2 - 4, 2015

Solar desalination management to fulfill greenhouse water demand using predictive control

L. Roca / CIEMAT, J.D. Álvarez, F. Rodríguez / UAL, **J. Bonilla** / CIEMAT - 2014

XXXV Jornadas de Automática 2014, Regular paper in Proceedings, Poster, Valencia (Spain), September 3 - 5, 2014

Intercambio de modelos entre herramientas de simulación con functional mock-up interface (FMI)

J. Bonilla / CIEMAT, J.D. Álvarez / US, A. de la Calle, L. Roca, L.J. Yebra / CIEMAT, F. Rodríguez / UAL - 2013

XXXIV Jornadas de Automática 2013, Regular paper in Proceedings, Poster, Terrassa (Spain), September 4 - 6, 2013.

Modelado y simulación de un evaporador de tubos sumergidos

A. de la Calle, L. Roca, **J. Bonilla**, L.J. Yebra / CIEMAT - 2011

XXXII Jornadas de Automática 2011, Regular paper in Proceedings, Poster, Seville (Spain), September 7 - 9, 2011

Sistema de ayuda a la toma de decisiones en la gestión del clima interior de un invernadero basado en modelos de crecimiento de cultivos de tomate

J. Bonilla, L.J. Yebra / CIEMAT, F. Rodríguez / UAL, Armando Ramírez / UACH - 2007

II Congreso Español de Informática (CEDI), I Simposio en Modelado y Simulación de Sistemas Dinámicos (SIMOSI), Regular paper in Proceedings, Oral presentation, Zaragoza (Spain), September 11 - 14, 2007.

International Doctoral Colloquia

Chattering problem in dynamic mathematical two-phase flow models

J. Bonilla / CIEMAT - 2009

5th Sollab Doctoral Colloquium on Solar Concentrating Technologies, Abstract, Oral presentation and Poster, Cologne, Germany, June 22 - 24, 2009

Chattering problem in dynamic mathematical two-phase flow models

J. Bonilla / CIEMAT - 2008

4th Sollab Doctoral Colloquium on Solar Concentrating Technologies, Abstract and Oral presentation, Tabernas, Almería (Spain), September 10 - 12, 2008

Research Stays

Object-oriented modeling and simulation of complex physical systems

2011

ETH Zurich, Department of Computer Science, Supervisor: Prof Dr. François E. Cellier, 3 months, Zurich (Switzerland), Sep 3 - Dec 3, 2011

Modeling of two-phase flow evaporators for parabolic-trough solar thermal power plants

2010

National Distance Learning University (UNED), Department of Computer Science and Automation, Supervisor: Prof Dr. Sebastián Dormido, 1 month, Madrid (Spain), Mar 1 - 31, 2010

Scientific Reports

I have contributed to more than 20 deliverables and scientific reports developed within funded international and national research projects, as well as for enterprises and public organizations. Selected representative examples are listed below.

Test THICAV - LEIA

2025

- **Authors**: J.A. Carballo **J. Bonilla**, A. Ávila-Marín, E. Orts / CIEMAT
- **Client**: Twer ingeniería.
- **Description**: Evaluación, análisis y modelado del redimiento del recepto THICAV de Tewer ingeniería, ensayos y validación con datos experimentales en la planta experimental CESA-I.

The potential until 2030 of concentrating solar power in Portugal using inductive projection planning

2021

- **Authors**: **J. Bonilla** / CIEMAT, G. Martín / Protermosolar.
- **Client**: European Solar Thermal Electricity Association (ESTELA).
- **Description**: Portugal electricity mix optimization for 2030 using the developed software tool based on artificial intelligence. Collaboration with Associação Portuguesa de Energias Renováveis (APREN) to establish the assumptions and goals considered in this study.

Optical model of concentrating systems offered to users

2011

- **Authors**: L.J. Yebra, **J. Bonilla**, L. Roca, E. Zarza / CIEMAT
- **Project**: SFERA: Solar Facilities for the European Research Area (228296)
- **Description**: Description of the optical model of concentrating systems offered to users for CIEMAT facilities in the scope of the SFERA project.

Membership to Scientific Committees

Joint Programme (tJP) on Digitalisation for Energy (DfE)

2021 - Now

Invited member of the tJP on Digitalization for energy. The European Energy Research Alliance (EERA), Subprogram 2: Data Science & Artificial Intelligence
Website

II Simposion Nacional de Ingeniería Hortícola (SNIH)

2016

Member of the organization commitee. Automatización y tecnologias de la información y comunicaciones (TICs) en la agricultura, University of Almería, February 10 - 12, 2016.

International Renewable Energy Congress (IREC)

2015 - Now

International program committee member. Institute of Electrical and Electronics Engineers (IEEE) Power & Energy Society.

Membership to Research Groups

Concentrating Solar Thermal Technologies

2026 - Now

Plataforma Solar de Almería (PSA), Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science, Innovation and Universities, Government of Spain

Point-Focus Concentrating Solar Thermal Technologies

2024 - 2025

Plataforma Solar de Almería (PSA), Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science, Innovation and Universities, Government of Spain

Solar Thermal Applications

2007 - 2023

Plataforma Solar de Almería (PSA), Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain

Automatic control, Robotics and Mechatronic (ARM) Research Group

2007 - Now

Department of Computer Sciences, University of Almería

Modeling and Control Unit

2007 - Now

Centro de Investigación en Energía Solar (CIESOL), Joint Institute UAL – PSA.CIEMAT

Membership to Strategic Research Lines

Analysis of sustainable energy systems

2021 - Now

Strategic Research Line, Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, April 2021

Plataforma Solar de Almería (PSA)

2021 - Now

Strategic Research Line, Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, April 2021

Experimental Campaigns

CESA-1 solar tower power plant

Singular scientific and technical infrastructure, Plataforma Solar de Almería (ICTS - PSA) - 2018 - Now

Experimental campaign to develop a new tracking strategy based on artificial intelligence and computer vision

NEP PolyTrough 1200 solar field

Singular scientific and technical infrastructure, Plataforma Solar de Almería (ICTS - PSA) - 2019

Experimental campaign for the validation of small parabolic trough collector models

AQUASOL Plant

Singular scientific and technical infrastructure, Plataforma Solar de Almería (ICTS - PSA) - 2018 - 2019

Experimental campaign to validate a steam generator model and the coupling of the elements of the AQUASOL plant

Heat Transfer Fluid (HTF) test loop

SFERA-II: Transnational access activities - 2016

Experimental campaign for the validation of parabolic trough collector models

Molten Salt Test Loop for Thermal Energy Systems (MOSA)

HYSOL project - 2016

Experimental campaign for the validation of heat exchanger and molten salt thermal storage system models

Prototypes and pilot plants

More efficient heliostat fields for solar tower plants (HELIOSUN)

2022 - 2025

Generation Knowledge Projects, Ministry of Science and Innovation, PID2021-126805OB-100, Work team member, budget: 145,200 €

Solar tower power mockup for the assessment of advanced control techniques

J.A. Carballo, **J. Bonilla** / CIEMAT, M. Berenguel / UAL, J. Fernández-Reche, G. García / CIEMAT - 2020

Renewable Energy, Volume 149, pages 682-690, ISSN: 0960-1481. DOI: 10.1016/j.renene.2019.12.075, 2020.
JIF: 8.001 (2020),
Q1 - Energy & fuels (16/144, 86.40% percentile)

Patents and Intellectual Property

Cámara con red neuronal integrada para caracterización geométrica y óptica de heliostatos (EN TRÁMITE)

J.A. Carballo, **J. Bonilla**, A. Ávila / CIEMAT, M. Berenguel / UAL - 2025

Patente, Unidad de Propiedad Industrial y Creación de Empresas de Base Tecnológica, Oficina de Transferencia de Tecnología del CIEMAT, no M-000601/2021, July 20, 2021.

Software basado en una red neuronal para caracterización geométrica y óptica de heliostatos (EN TRÁMITE)

J.A. Carballo, **J. Bonilla**, A. Ávila / CIEMAT, M. Berenguel / UAL - 2025

Propiedad intelectual, Unidad de Propiedad Industrial y Creación de Empresas de Base Tecnológica, Oficina de Transferencia de Tecnología del CIEMAT, no M-000601/2021, July 20, 2021.

Artificial intelligence software tool for the optimization of the electricity mix

J. Bonilla - 2021

Propiedad intelectual, Unidad de Propiedad Industrial y Creación de Empresas de Base Tecnológica, Oficina de Transferencia de Tecnología del CIEMAT, no M-000601/2021, July 20, 2021.

Technology Transfer

Simulación para la Optimización del Mix Eléctrico Portugués en 2030

2021

Contrato de prestanción de servicios para la realización de trabajos de investigación entre la Asociación Española para la Promoción de la Industria Termosolar (PROTERMOSOLAR) y Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas (CIEMAT). Budget: 5,000 €.

Diseño de modelos y estrategias de simulación, control, ensayo y monitorización del lazo Senertrough

2007

Contrato de transferencia de investigación para Cobra instalaciones y servicios S.A. Modelo y simulador de planta de captadores cilindroprabólico de la planta Andasol. Validación con datos de campañas experimentales. Budget: 337,982 €.

Research Project Reviewer and Peer-review Activities

R&D international project reviewer

2023

Revisor para proyectos nacionales en Paraguay. Fomento a la investigación científica en el marco del programa paraguayo para el desarrollo de la ciencia y tecnología (PROCIENCIA II). Programa Iberoamericano de Ciencia y Tecnología para el Desarrollo (CYTED), Consejo Nacional de Ciencia y Tecnología (CONACYT), Paraguay.

Regular reviews for scientific journals

2012 - Now

Nature Energy, Solar Energy, Renewable Energy, Energy Conversion and Management, Energy, Energies, Applied Thermal Engineering, International Journal of Heat and Mass Transfer, etc.

Regular reviews for scientific conferences

2012 - Now

Concentrating Solar Power and Chemical Energy Systems (SolarPaces), International Renewable Energy Congress (IREC), International Conference on Sustainable Energy Information Technology, European Control Conference (ECC), Annual Conference of the IEEE Industrial Electronics Society (IECON), International Conference on Mathematical Modelling (MATHMOD), etc.

Host of Facility Access Activities

SFERA-II: Transnational access activities

2016

Access to Plataforma Solar de Almería (PSA) facilities. Collaboration for the experimental validation of a Modelica-based dynamic parabolic trough collector model, A. Desideri, R. Dickes, S. Quoilin / University of Liege (Belgium), 3 weeks, Jun 27 - Jul 8, 2016

Outreach Activities

Plataforma Solar de Almería: Reutiliza, desala y reduce

2025

Open Researchers 2024-25, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 26, 2025

Cambio Climático y Energía Solar

2025

pequeCIEMAT, CIEMAT outreach activities, 5º y 6ª primaria, 90 students, CEIP Freinet, Almería (Spain), May 27, 2025

Certificate of participation in science outreach activities

2024

Unidad de Divulgación Científica (UDC), CIEMAT

La ciencia del Sol: Explorando la energía termosolar y el almacenamiento de calor

2024

Open Researchers 2023-24, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 27, 2024

Plataforma Solar de Almería: La Energía Termosolar Clave Contra El Cambio Climático

2023

European Researchers' Night, open researchers, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 29, 2023

Energía Solar Térmica e Inteligencia Artificial

2023

SOL-préndete: Didáctica y divulgación de la energía solar térmica de concentración con nuevas tecnologías de realidad aumentada y virtual, 1º bachillerato science students, IES Sol de Portocarrero, Almería (Spain), June 16, 2023

Energía Solar Térmica e Inteligencia Artificial

2023

SOL-préndete: Didáctica y divulgación de la energía solar térmica de concentración con nuevas tecnologías de realidad aumentada y virtual, 4º ESO science, humanities and social sciencie students, IES Cerro Milano, Almería (Spain), February 22, 2023

SOL-préndete: Didáctica y divulgación de la energía solar térmica de concentración con nuevas tecnologías de realidad aumentada y virtual

2022

European Researchers' Night, open researchers, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 30, 2022

Inteligencia artificial y energía solar

2019

Foco Sur - Actualidad, Economía y Cultura de Almería, no 262, December 2019

Heliostato inteligente

2019

European Researchers' Night, open researchers, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 27, 2019

Modelado y control de sistemas de concentración solar

2018

European Researchers' Night, open researchers, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 28, 2018

Modelado, simulación y control de sistemas solares

2017

European Researchers' Night, open researchers, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 29, 2017

Modelado, simulación y control de sistemas solares

2016

European Researchers' Night, open researchers, Marie Skłodowska-Curie actions, European Commission, Almería (Spain), September 30, 2016

Control y operación automático de plantas termosolares

J. González / Aunergy, M. Berenguel / UAL, M. Romero, L.J. Yebra, D. Martínez, A. Valverde, L. Roca, **J. Bonilla**, E. Zarza / CIEMAT - 2009

Montajes e Instalaciones, Year no 39, No 437, March 2009, pages 90-95, ISSN: 0210-184X

Press News

CST4ALL Newsletter No. 4

2025

Support to the activities of the concentrated solar thermal technology area of the SET plan

New AI perfects heliostat aim to boost solar tower power

2025

SolarPACES

Investigadores de Andalucía diseñan un sistema basado en IA para "mejorar el rendimiento de plantas solares"

2024

Europa Press

Investigadores Andaluces Desarrollan Sistema de IA para Optimizar el Rendimiento de Plantas Solares

2024

Andalucía Informa

Diseñan un sistema ‘inteligente’ para mejorar el rendimiento de las plantas solares

2024

Fundación Descubre

¿Cómo puede una IA aumentar la energía capturada en plantas solares?

2024

Review Energy Spain

Diseñan un sistema basado en IA para mejorar el rendimiento de plantas solares

2024

El periódico de la energía

Investigadores de Almería diseñan un sistema de IA para mejorar el rendimiento de plantas solares

2024

IDEAL

Investigadores andaluces diseñan un sistema basado en IA para “mejorar el rendimiento de plantas solares”

2024

La Vanguardia

Cómo puede la Inteligencia Artificial mejorar mejorar el rendimiento de las plantas solares fotovoltaicas

2024

Nova Ciencia

Desarrollan un sistema de IA en Andalucía para optimizar plantas solares

2024

Hora Punta

Desarrollan un sistema de IA en Andalucía que aumenta la eficiencia de plantas solares

2024

Nuevo Diario

Investigadores andaluces diseñan un sistema basado en IA para "mejorar el rendimiento de plantas solares"
2024

MSN

Diseñan en Andalucía un sistema basado en IA para mejorar el rendimiento de plantas solares
2024

Az Costa del Sol

Investigadores de la Plataforma Solar de Almería mejoran el rendimiento solar con IA
2024

Noticias de Almería

Desarrollan un sistema de IA para optimizar plantas solares en Andalucía
2024

IY Magazine

Investigadores andaluces diseñan un sistema basado en IA para "mejorar el rendimiento de plantas solares"
2024

Gente Digital

Investigadores andaluces usan IA para mejorar rendimiento de plantas solares
2024

Córdoba BN

Diseñan un sistema de IA para mejorar el rendimiento de plantas solares
2024

Almería Información

Mejoran el rendimiento de las plantas solares con inteligencia artificial
2024

CuentameAlgoBueno

Investigadores andaluces diseñan un sistema basado en IA para "mejorar el rendimiento de plantas solares"
2024

Teleprensa

Noticias, sala de prensa, CIEMAT
2016 - Now

- Participación de la PSA-CIEMAT en el CIES 2020 celebrado en Lisboa
- Software para optimización del mix eléctrico aplicando técnicas de inteligencia artificial
- Tesis doctoral del proyecto ENERPRO de un investigador de CIESOL, Centro mixto PSA-UAL
- Seis Premios Extraordinarios de Doctorado en la Plataforma Solar de Almería
- Finaliza con éxito el proyecto HYSOL

Scientific Software

Weather station software
J. Bonilla, J.A. Carballo

A real-time and modular weather station software architecture based on microservices which offers a scalable, flexible, and easily maintainable solution for weather data collection and dissemination

CSP Data
J. Bonilla

Detailed information with filtering and exporting capabilities about all commercial concentrating solar thermal power plants around the world

Electricity mix optimization

J. Bonilla, J. Blanco, E. Zarza, D.C. Alarcón-Padilla

Spanish mix optimizations for 2030, 2040 and 2050 applying Artificial Intelligence

Hel-IoT web server: a smart heliostat development platform

J.A. Carballo, **J. Bonilla**, J. Fernández-Reche, D.C. Alarcón-Padilla - 2022

28th Concentrating Solar Power and Chemical Energy Systems (SolarPaces)
Abstract, Poster, DOI: 10.52825/solarpaces.v1i.721, Albuquerque (USA), Sep 27 - 30, 2022.

Parabolic trough collector (PTC) power plant performance

J. Bonilla, E. Zarza

Web app to estimate the performance of a 50 MW PTC power plant with 8 hours of thermal storage in different locations around the world

Solar tower, gas turbine and MED-TVC simulator

J. Bonilla, P. Palenzuela, B. Ortega-Delgado, J. Fernández-Reche, D.C. Alarcón-Padilla

Web app for the integration of a thermal seawater desalination system, based on Multi-Effect Distillation technology with Thermal Vapor Compression (MED-TVC), into a high-temperature power cycle (air Brayton cycle) and high-temperature Concentrating Solar Power (CSP) technology, Central Receiver Solar (CRS) tower system.

Surf - Simulator builder

J. Bonilla

A toolchain for building powerful multi-platform simulators of FMI-compliant models in an easy and convenient way. Developed simulators:
- Small Parabolic Trough Field
- Thermal Energy Storage Tank
- Thermodynamic Properties of Solar Fluids
- Big Parabolic Trough Collector
- Heat Transfer Basics

Optifmus - Optimization of functional mock-up units (FMUs)

J. Bonilla

Open source multi-platform software tool for optimization studies in FMI-compliant models applying genetic algorithms

Contributions to Scientific Open Source Software

SolarPILOT and CoPylot

2022

Solar Power tower Integrated Layout and Optimization Tool (SolarPILOT) and library to access all capabilities seamlessly through Python (CoPylot), National Renewable Energy Laboratory (NREL) (USA). - **Contribution**: code to read SolarPILOT (spt) files and import them in CoPylot.

Flutter web ChartJS

2020

Charting components for Flutter Web, Victor Ramos / Fliper (Brazil). - **Contribution**: linear, logarithm and time axes for charts.

ThermoCycle library

2018

A Modelica library for the simulation of thermodynamic systems, University of Liège (Belgium) and DTU Mechanical Engineering (Denmark). - **Contribution**: improvements to the solar collector models.

Main International Collaborations

Machine learning for solar thermal systems

2018 - Now

Pete Warden & Aakanksha Chowdhery. TensorFlow Mobile/Embedded Team, Google Brain, USA

Object-oriented modeling and simulation of complex physical systems

2011 - 2015

François E. Cellier. Department of Computer Science, ETH Zurich, Switzerland.

Related publications:
- Switching moving boundary models for two-phase flow evaporators and condensers
- Object-oriented library of switching moving boundary models for two-phase flow evaporators and condensers
- Object-oriented modeling of switching moving boundary models for two-phase flow evaporators

Cloud detection in images applying artificial intelligence

2025

Bijan Nouri, Department of Qualification, The German Aerospace Center (DLR), Germany.

Related publications:
- Cloud Detection and Tracking Based on Object Detection with Convolutional Neural Networks

Modeling, optimization and control of solar thermal systems

2016 - Now

Alberto de la Calle. Solar Energy System Group, Commonwealth Scientific and Industrial Research Organisation (CSIRO), Australia.

Related publications:
- Optimal operation of solar thermal desalination systems coupled to double-effect absorption heat pumps
- Design and experimental validation of a computational effective dynamic thermal energy storage tank model
- Optimal operating conditions analysis for a multi-effect distillation plant according to energetic and exergetic criteria
- Optimal operating conditions analysis of a multi-effect distillation plant - Study on shell-and-tube heat exchanger models with different degree of complexity for process simulation and control design
- Modelo dinámico de un recuperador de gases - sales fundidas para una planta termosolar híbrida de energías renovables
- Dynamic modeling and simulation of a double-effect absorption heat pump
- Predictive control applied to a solar desalination plant connected to a greenhouse with daily variation of irrigation water demand

Dynamic modeling, validation and optimization of solar thermal systems

2016 - 2018

Sylvain Quoilin, Adriano Desideri & Rémi Dickes. Aerospace and Mechanical Engineering Department, University of Liege, Belgium.

Related publications:
- Steady-state and dynamic validation of a parabolic trough collector model using the ThermoCycle Modelica library

Data Discovery Web Application of Commercial CSP Plants (CSP Data)

2022 - Now

Johan Lilliestam, Sustainability Transition Policy Friedrich-alexander-universität Erlangen-Nürnberg, Germany.

Related publications:
- CSP Data: A Data Discovery Web Application of Commercial CSP Plants

Data Discovery Web Application of Commercial CSP Plants (CSP Data)

2022 - Now

Richard Thonig, Research Institute For Sustainability Helmholtz Centre Potsdam, Germany.

Related publications:
- CSP Data: A Data Discovery Web Application of Commercial CSP Plants

New design methods for heat exchangers

2018

Reza Zarghami, Ahad Mohammadi & Shahab Golshan. Process Design and Simulation Research Centre, School of Chemical Engineering, College of Engineering, University of Tehran, Iran.

Related publications:
- A novel heat exchanger design method using a delayed rejection adaptive Metropolis Hasting algorithm

Direction of Ph.D. Students (2)

Modelado y optimización para una gestión eficiente de recursos en tecnlogía termosolar

Jose Antonio Carballo, University of Almería (UAL) - 2018

Directors: **J. Bonilla** / CIEMAT and M. Berenguel / UAL
Ph.D. thesis

Contribuciones al modelado dinámico de procesos termoquímicos en instalaciones termosolares

Alberto de la Calle Alonso, National Distance Learning University (UNED) - 2015

Directors: L. Roca, **J. Bonilla** / CIEMAT and S. Dormido / UNED
Ph.D. thesis

Direction of Final Degree Projects

Diseño, montaje y control de una maqueta de planta solar de receptor central inteligente

Luis Ignacio Jiménez Martínez, University of Almería (UAL) - 2025

Directors: M. Berenguel / UAL and **J. Bonilla** / CIEMAT
Final degree project

Desarrollo de una herramienta software para simulación y control del demostrador de una planta termosolar híbrida de energías renovables

José Martín de Dios, University of Almería (UAL) - 2016

Directors: J.D. Álvarez / UAL and **J. Bonilla** / CIEMAT
Final degree project

Staff Supervision

Company tutor

Álvaro Martínez Fernández - 2022 - 2023

Curricular Practices, code 356971, Centro de Investigación en Energía Solar (CIESOL), University of Almería (UAL)

Director of research plan

Jose Antonio Carballo - 2015 - 2018

Ph.D. in Computer Science, code 8909, Real Decreto 99/2011 de 28 de enero (BOE nº 35, de 10 de febrero de 2011), University of Almería (UAL)

Continuing Education (28)

React
2023

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, CIEMAT in-company training, Online, course, 20 hours. Almería, September 18 - 21, 2023.

Sistemas Solares Térmicos de Media Concentración
2023

Dr. Eduardo Zarza, Plataforma Solar de Almería, 12 hours, May 15, 17, 18 and 23, 2023

La transición energética - La energía solar en el sistema eléctrico español
2022

IEEE Power & Energy Society (PES), Webinar 11, Spain Section Chapter, PE31, 3 hours, May 19, 2022.

Neural networks applied to scientific and technical problems
2021

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, CIEMAT in-company training, Online, course, 50 hours. Almería, October 4 - November 5, 2021.

Design thinking and creativity techniques
2021

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, CIEMAT in-company training, Online, course, 5 hours. Almería, May 24 - 25, 2021.

Introduction to applied supervised machine learning
2020

Dr. Antonio Jesús Fernández (Universidad de Extremadura), Universidad of Almería, Online, course, 10 hours. Almería, May 26 - 28, 2020

Jornada sobre Transición del sector eléctrico español hacia un mercado mayoritariamente renovable en 2030
2019

Máster Unversitario en Energía Solar, Universidad of Almería, Almería, Presential, course, 5 hours, 16 de mayo de 2019.

Practical introduction to quantum computing
2019

Dr. Elías Fernández-Combarro Álvarez (Universidad de Oviedo), Universidad of Almería, Presential, course, 10 hours. Almería, October 1 - 2, 2019

Industry 4.0
2018

Dr. Ramón Vilanova (Universidad Autónoma de Barcelona), Universidad of Almería, Presential, course, 10 hours. Almería, November 14 - 15, 2018

Advanced control techniques for process control
2018

Dr. Antonio Visioli (University of Brescia), Universidad of Almería, Presential, course, 10 hours. Almería, November 25 - 26, 2018

Programming of GPU cards with CUDA
2017

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Economy, Industry and Competitiveness, Government of Spain, CIEMAT in-company training, Online, course, 15 hours.

Almería, October 16 - 20, 2017

Introduction to acceleration and parallelization of scientific code

2017

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Economy, Industry and Competitiveness, Government of Spain, CIEMAT in-company training, Online, course, 12 hours. Almería, October 9 - 12, 2017

Confort y Edificación Sostenible, una respuesta desde la Automática y las Micro-Redes Energéticas

2016

Parque Científico-Tecnológico de Almería (PITA), Presential, course, 5 hours. Almería, Oct 20, 2016

Mathematica and SystemModeler

2016

Wolfram Research, Universidad of Almería, Presential, course, 8 hours. Almería, May 16 - 17, 2016

Optimization techniques applied to multi-objetive problems: Evolutionary Algorithms

2012

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Economy and Competitiveness, Government of Spain, CIEMAT in-company training, Online, course, 12 hours. Almería, October 2 - 10, 2012

Introduction to Python programming

2012

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Economy and Competitiveness, Government of Spain, CIEMAT in-company training, Online, course, 12 hours. Almería, June 25 - July 6, 2012

Introduction to the MPI and OpenMP programming models in the Euler supercomputer

2011

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, CIEMAT in-company training, Online, course, course, 10 hours. Almería, June 13 - 17, 2011

Engineering equation solver, CIESOL

2009

Centro Mixto UAL-PSA.CIEMAT, conducted by Plataforma Solar de Almería (PSA), Presential, seminar, 16 hours. Almería, December 1 - 2, 2009

Efficient presentations

2009

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, CIEMAT in-company training, Online, course, 20 hours. Almería, November 23 - December 18, 2009

Educational technology: Remote virtual laboratories, interactive tools and Microsoft collaborative technologies

2009

University of Almería, coducted by Prof. Dr. Manuel Berenguel Soria, Vicar (Almería), Presential, course, 25 hours. Vicar (Almería), July 13 - 16, 2009

Occupational health & safety management systems at the Plataforma Solar de Almería

2008

Centro de Investigaciónes Energéticas, Medioambientales y Tecnológicas (CIEMAT), Ministry of Science and Innovation, Government of Spain, Almería, Online, course, 9.5 hours. Almería, October 27 - 31, 2008

Identification of dynamic systems

2008

University of Almería, conducted by Prof. Dr. Daniel E. Rivera, Arizona State University (EEUU), Almeria, Presential, course, 15 hours. Almeria, September 8 - 10, 2008

The influence of IT in the learning process: a practical point of view

2008

University of Almería, coducted by Prof. Dr. Manuel Berenguel Soria, Vicar (Almería), Pressential, course, 25 hours. Vicar (Almería), July 16 - 18, 2008

Process control: application in the Matlab-Simulink environment

2007

National Distance Learning University (UNED), coducted by Prof. Dr. Sebastián Dormido Bencomo, El Barco de Ávila (Ávila), Presential, course, 35 hours. El Barco de Ávila (Ávila), July 16 - 20, 2007

Modelica and OpenModelica

2007

National Distance Learning University (UNED), Departamento de Informática y Automática, conducted by Prof. Dr. Peter Fritzson, Linköping University (Sweden), Madrid, Presential, course, 25 hours. Madrid, March 26 - 28, 2007

LaTeX: a scientific editing platform

2007

University of Almería, Enseñanzas propias, Almería, Presential, course, 40 hours. Almería, March 8 - 30, 2007

Programming languages: Java and Visual Basic

2004

The Andalusian Government, Employment and technological development regional Ministry, Info-Studio Academy, Almería, occupational formation course, 400 hours. Almería, January 8 - May 3, 2004

Legal and technical issues of security on the Internet. An approach to safe e-commerce

2000

International university of Andalusia, University of Almería, Almería, Presential, course, 30 hours. Almería, July 24 - 28, 2000

Prizes, Acknowledgments and Fellowships

4ª Edición De Los Premios del Mar y de la Economía Azul Sostenible de Andalucía (PROA)

2023

Galardón en la categoría de Conocimiento Azul concedido a la la Unidad de Aplicaciones Solares Térmicas de la Plataforma Solar de Almería. Patrocinados por la Consejería de Sostenibilidad, Medio Ambiente y Economía Azul y cuentan con la colaboración del Museo de Almería, que depende de la Consejería de Turismo, Cultura y Deporte de la Junta de Andalucía, el Campus de Excelencia Internacional del Mar (CEI·MAR) y Cocom Media. Viernes, 15 de diciembre de 2023, Museo de Almería.

Premio al Fomento de la Investigación Aplicada a la Empresa (XI Edición)

2018

Galardón condecido al Grupo de Automática, Robótica y Mecatrónica en la categoría “Premios a los grupos de investigación de la Universidad de Almería que se distingan especialmente por sus actividades de investigación contratadas con empresas e instituciones” otorgado por el Consejo Social de la Universidad de Almería.
Video entrega del galardón

Ph.D. dissertation extraordinary prize in Engineering

2016

Modeling of Two-Phase Flow Evaporators for Parabolic-Trough Solar Thermal Power Plants, University of Almería (UAL), **J. Bonilla**, Director: L.J. Yebra and S. Dormido

Director of Ph.D. dissertation with an extraordinary prize

2016

Contribuciones al modelado dinámico de procesos termoquímicos en instalaciones termosolares, National Distance Learning University (UNED), A. de la Calle, Director: L. Roca, **J. Bonilla** and S. Dormido

Best paper award given by sociedad española de agroingeniería

2016

Control predictivo para satisfacer la demanda de agua en un invernadero mediante un sistema de desalación solar, II Simposio Nacional de Ingeniería Hortícola - Automatización y TICs en Agricultura

Best paper in modeling and simulation

2013

Intercambio de Modelos entre Herramientas de Simulación con Functional Mock-up Interface (FMI), XXXIV Jornadas de Automática 2013, Comité Español de Automática (CEA)

Ph.D. scholarship

2007 - 2009

Plataforma Solar de Almería, Energy Deparment, Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas (CIEMAT), Spain

Undergraduate scholarship

2004

University of Almería - Plataforma Solar de Almería agreement, Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas (CIEMAT), Spain